

Application of saffron as a neuroprotective agent

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28.1 Alzheimer's disease

28.1.1 Introduction and epidemiology

Alzheimer's disease (AD) is one of the most common neurodegenerative disorders known in the world. This condition usually presents with symptoms during old age. It is estimated that approximately 5 million people are living with AD in the United States and by 2050 this number is expected to exceed over 13.8 million Americans (Hebert et al., 2013; Kinney et al., 2018). Advancing age is one of the most important risk factors; thus, it is believed that after age 65, the risk for developing AD increases twofold every 5 years. Females are more likely to suffer from AD than males and in some studies family history of AD was reported (Jönsson et al., 2006).

AD has a significant financial burden on communities. According to some studies, the financial burden inflicted by AD is approximately \$230 billion, a number that is expected to increase over \$1.1 trillion by 2050 (Kinney et al., 2018). One study showed that there is a strong relationship between costs and cognitive changes (Jönsson et al., 2006). In addition, patients with severe AD often depend on others and require extensive caregiver time (Herrmann and Gauthier, 2008).

28.1.2 Signs and symptoms

Cognitive changes in AD follow a defined pattern. Its symptoms usually begin with memory problems. Subsequently, other brain functions are influenced and deterioration of virtually all intellectual functions, increasing apathy, and decreasing speech function gradually appear (Wolinsky et al., 2018). Tests are used to detect and measure memory and cognitive dysfunction such as the Alzheimer's disease assessment scale cognitive subscale (ADAS-cog), clinical dementia rating scale sums of boxes (CDR-SOB), and mini-mental state examination (MMSE) (Hughes et al., 1982).

28.1.3 Pathogenesis

Although there is no certain mechanism leading to Alzheimer's, two theories are commonly presented as major explanations for the pathogenesis of AD (Kinney et al., 2018). The first mechanism is the presence of the misfolded amyloid

plaques in the brain. When amyloid precursor protein is cleaved improperly by λ -secretase and erroneous β secretase, insoluble amyloid plaques aggregate in the brain. This theory arises from detection of certain gene mutations, which lead to significant increases in incidence of AD (Kinney et al., 2018; Wolinsky et al., 2018). The second important pathway leading to AD is hyperphosphorylation of tau, a protein that stabilizes microtubules in neurons. In addition, several studies have reported the significant role of uncontrolled brain inflammation behind AD progression. A number of examinations have shown high levels of inflammatory cytokines and macrophages in the histology of AD patient brains (Kinney et al., 2018). It is believed that in AD patients, the balance of neurotransmitter systems are impaired especially in glutaminergic and cholinergic systems. Excessive glutamate is released, which leads to cellular damage and cell death (Waite, 2015; Wolinsky et al., 2018).

28.1.4 Pharmacological therapy

There are many pharmacological therapies to control cognitive dysfunction of AD; however, none are curative (Waite, 2015). Five drugs have been developed and used in several clinical trials: four acetylcholinesterase inhibitors (AChEIs) (donepezil, rivastigmine, galantamine, tacrine) and one *N*-methyl *D*-aspartate (NMDA) glutamate receptor blocker (memantine) (Waite, 2015; Wolinsky et al., 2018). Donepezil is approved to control mild-to-moderate AD. In one study, administration of donepezil over 6 months resulted in an improvement of 1.4 points on the MMSE (Waite, 2015). However, usually AChEIs are accompanied by nausea, vomiting, sleepless, weakness, and muscle cramps (Wolinsky et al., 2018). Memantine is another drug that has a significant effect on cognition and behavior and it is approved for moderate-to-severe Alzheimer's (Herrmann and Gauthier, 2008). There are some alternative medications to alleviate symptoms of AD like antipsychotic drugs such as risperidone and antidepressant drugs such as citalopram, although they do not have a significant role in controlling symptoms (Wolinsky et al., 2018). Several new studies recommend novel strategies for AD therapy. One of the most important of these strategies is to decrease β amyloid aggregation in the brain by targeting the secretase enzyme (Waite, 2015).

28.2 Saffron in Alzheimer's disease treatment

28.2.1 Introduction to saffron

Herbal medicine still has an important role in primary health care for approximately 80% of the world's population in both developing and developed countries due to better cultural acceptability and safety profiles (Akhondzadeh and Abbasi, 2006). Saffron is a spice derived from the flower of *Crocus sativus*. It is usually used as a nutritional supplement, and it is found all over the world. It is estimated that Iran supplies 90% of the world's production of saffron (Christodoulou et al., 2015). Saffron includes many components like carbohydrates, polypeptides, lipids, and H_2O ; however, the main components with medical effects are crocetin, crocin, picrocrocin, and safranal (Khorasany and Hosseinzadeh, 2016).

Saffron has many effects on the neurotransmitter network in the brain. Some limited investigations have been designed to assess the relationship between saffron and cholinergic activity. In one study, the inhibitory effect of aqueous methanolic saffron extract on AChE was reported (Geromichalos et al., 2012). Crocetin was found to bind to the catalytic and anionic site of AChE. This result demonstrated the potential effect of saffron to increase cholinergic activity. There have also been studies to assess the effects of saffron and its constituents on the glutamate system by focusing on NMDA receptors. The result of one study demonstrated that saffron extracts and crocetin had clear binding capacity at the phencyclidine binding site of the NMDA receptor and at the $\sigma 1$ receptor (Hensel et al., 2008). The results also revealed that crocins and picrocrocin were not effective agonists. In an animal study, subchronic administration of ketamine, a noncompetitive NMDA receptor antagonist, caused impairment in cognitive function in rats and a single injection of crocins reversed recognition memory deficits produced by ketamine in rats (Georgiadou et al., 2014). It is believed that saffron can also increase reuptake inhibition of dopamine and norepinephrine (Alavizadeh and Hosseinzadeh, 2014).

28.2.2 Effects of saffron on the central nervous system

Effects of saffron and its derivatives, like safranal, crocin, and crocetin, on the central nervous system (CNS) have been widely studied and various benefits have been elicited.

Crocine, the active constituent of saffron, protects against beryllium chloride toxicity in rats through diminution of oxidative stress and increased gene expression of antioxidant enzymes (Lopresti and Drummond, 2014). It also attenuates lipid peroxidation and protects against tissue injury (Altinoz et al., 2016). Oral crocine may have protective effects against intracerebroventricular streptozotocin-induced spatial memory deficit and oxidative stress in rats (Shafiee et al., 2018). In one study, it was shown that crocine can attenuate acute hypobaric hypoxia-induced cognitive deficits in rats (Zhang et al., 2018).

Safranal is an active ingredient in saffron. Safranal attenuated cerebral ischemia induced oxidative damage in rat hippocampus. There is an anticonvulsant property in safranal and its effect may be due to modifications on the benzodiazepine binding sites of the gamma-aminobutyric acid A (GABA_A) receptor complex. Safranal can be effective in protecting a susceptible aged brain from oxidative damage by increasing antioxidant defenses (Samarghandian et al., 2016).

Crocine potentiates neurite growth in hippocampal neurons and facilitates functional (both motor and sensorimotor) recovery in rats with spinal cord injury (Wang et al., 2017a). Crocine effectively attenuates neuropathic pain and significantly suppresses oxidative stress and neuroinflammation in the spared nerve injury (SNI) mouse model, supporting crocine potential in treatment against neuropathic pain (Wang et al., 2017b). Crocine attenuates spatial learning dysfunction and hippocampal injury in a model of vascular dementia (Lopresti and Drummond, 2014).

Oxidative stress has an important role in progression of a number of neurodegenerative diseases, including AD. In several studies, a close relationship was observed between cognitive impairment and plasma antioxidant capacity of patients, and it is believed that progression of cognitive dysfunction is related to degree of oxidative stress (Perry et al., 2002). Studies have shown that saffron may play a role in attenuating age-related oxidative damage in the rat hippocampus (Samarghandian et al., 2016). The effect of saffron ethanol extract on inhibition of experimental autoimmune encephalomyelitis in C57Bl/6 mice has also been studied and the results suggest for the first time that saffron is effective in prevention of symptomatic experimental autoimmune encephalomyelitis (EAE) by inhibition of oxidative stress and leukocyte infiltration to the CNS and may be potentially useful for the treatment of multiple sclerosis (Ghazavi et al., 2009). Crocine, one of the major component of saffron, was found as a potent neuronal antioxidant (Hosseinzadeh et al., 2009). In one study, crocine was reported as a unique and potent antioxidant in neurons, and it was observed that it can protect neuronally differentiated pheochromocytoma cells (PC-12) from peroxidation of their cell membrane lipids and decrease superoxide dismutase activity in the neurons (Ochiai et al., 2004). It has also been reported that crocine has a more antioxidant effect than α -tocopherol. In one animal study, there was observed that administration of crocine was effective to protect oligodendrocytes in the brain of the mouse (Nam et al., 2010). These studies confirm that crocine is an important agent to suppress oxidative stress and exhibits neuroprotective activities.

The mechanism(s) underlying the effects of saffron and its active constituents on cognition is still a matter of investigation. Among the potential mechanisms, promotion of long-term potentiation, anti-amyloidogenic activity, inhibition of AChE activity, and potent antioxidant properties are proposed to explain some of saffron effects on cognition. Long-term potentiation in the hippocampus is a form of activity-dependent synaptic plasticity and is believed to be a cellular basis for learning and memory. Reportedly, in this context, it was demonstrated that crocine promoted hippocampal long-term potentiation. *In vitro* enzymatic and molecular docking studies indicate that saffron and crocine, but not safranal, inhibit AChE by binding to the catalytic center and the peripheral anionic sites. This AChE inhibitory activity displayed by saffron and crocine promotes an increase in synaptic acetylcholine levels, a critical neurotransmitter in cognitive functions (Pitsikas, 2015).

As mentioned previously, amyloid- β (A β) peptide and tau proteins play an important role in AD and new therapeutic strategies are focused on these targets. Aggregation of β amyloid can cause neuron cell apoptosis and synaptic dysfunction. Several studies have demonstrated inhibitory effects of saffron extract and crocine on fibrillogenesis of A β . One investigation revealed that saffron extract tightens the blood–brain barrier and reduces amyloid β load and related toxicity in 5XFAD mice (Batarseh et al., 2017). One study exhibited inhibitory effects of saffron and crocine on fibrillogenesis of A β in a concentration and time-dependent manner (Papandreou et al., 2006); these results were also reported in other studies (Ghahghaei et al., 2012). Another investigation studied effects of crocine on fibrillogenesis of A β by thioflavine T-based fluorescence assay, DNA binding shift assay, CD spectroscopy, and transmission electron microscopy. Its results showed that crocine, through stabilization of the helical structure, could prevent A β -mediated amyloid fibril formation *in vitro* (Ghahghaei et al., 2013). Some studies were designed to measure crocine's neuroprotective activity against A β -induced toxicity in neuronal tissue (Morelli et al., 2016). Administration of A β produced significant damage to cells through induction of reactive oxidative species and apoptosis. The results showed that when A β was administered together with crocine, a significant dose-dependent inhibition of A β -induced apoptosis and reactive oxygen species production was observed. This may also suggest that crocine has the potential to prevent the aggregation of A β peptide and its adverse effects. Another important mechanism of AD is the abnormal accumulation of

hyperphosphorylated tau protein and intracellular aggregation of microtubules-associated tau protein. It is believed that assembled microtubules and stabilization of the microtubule network plays an important role in signal transmission. Impaired phosphorylation of tau causes formation of neurofibrillary tangles, which leads to damaged neurons. Inhibition of abnormal hyperphosphorylation may provide a wide therapeutic target for AD (Iqbal et al., 2010). The inhibitory effects of crocin on aggregation of hyperphosphorylated tau was evaluated in one study using both biochemical methods and cell culture (Karakani et al., 2015). The results show that administration of crocin stabilizes the system by preventing fibrillation and lowering the tendency for aggregation, which was confirmed by transmission electron microscopy images. The results proved that crocin provided neuroprotection by reducing acrolein-induced A β and tau phosphorylation.

28.2.3 Neuroprotective activity of saffron

Extensive studies have shown the impact of oxidative stress on the cognitive activity of neurons. Increased production of reactive oxygen species or reduced activity of antioxidant enzymes can lead to increased oxidative stress (Ghadrdoost et al., 2011). According to some studies, pathological situations such as stroke, trauma, or AD can deprive neurons of oxygen and lead to apoptosis. Oxidative stress is one of the mainstays in the process of cellular apoptosis and it appears that controlling it can be an important step in increasing the lifespan of neuronal cells (Soeda et al., 2016). The crucial point is the relevance of the level of these free radicals to the severity of cognitive impairments such as AD. In an animal study, crocin demonstrated its protective effect on oligodendrocytes of the mouse brain by suppressing the stress of the endoplasmic reticulum and improving the behavior of the animal (Deslauriers et al., 2011). In addition, the inhibitory effect of crocin on proapoptotic factors such as bcl-2 and Bax and Caspase-3 has been demonstrated in PC-12 cells damaged by H₂O₂ (Soeda et al., 2016). Crocin seems to be an effective agent in protecting the hippocampus. In an animal study, where streptozotocin induced diabetes on mice, administering of crocin resulted in protection of the hippocampal neurons, improving learning activity and increasing antioxidant capacity (Tamaddonfard et al., 2013). In another study, the possible effect of saffron on the hypothalamus-pituitary-adrenal axis was debated and its protective role in improving cognitive activity was shown (Ghadrdoost et al., 2011). Saffron can have effective effects on motor activity. In an animal study where mouse dopaminergic neurons were eliminated by 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine, administering saffron reduced the damage to neurons by preserving neurotrophic factors such as brain-derived neurotrophic factor (BDNF) (Haeri et al., 2019).

All of these studies have clearly shown the important role of saffron in protecting brain cells and point out that this substance can be a potential factor in the improvement of cognitive brain disorders.

28.2.4 Saffron clinical trials on Alzheimer's disease

Saffron was administered in a few clinical trials and found to be comparable to AChEI drugs and NMDA receptor blockers.

In a multicenter, randomized, double-blind controlled clinical study, saffron (30 mg day⁻¹ or 15 mg twice daily) or donepezil (10 mg day⁻¹) was administered to 54 patients aged 55 years and older with mild-to-moderate AD (Akhondzadeh et al., 2010b). The patients were followed over 22 weeks and were evaluated by ADAS-cog and CDR-SOB. Saffron was found to be as effective as donepezil in this clinical trial. There was no significant difference between the two groups in frequency of side effects; however, patients in the saffron extract group experienced less vomiting.

In another 16-week trial, saffron was compared with placebo (Akhondzadeh et al., 2010a). Forty-six patients with mild-to-moderate AD were randomly assigned to two groups to receive saffron (30 mg day⁻¹ or 15 mg per capsule, twice per day) or placebo (two capsules per day). Cognitive status and clinical profiles of patients were assessed by ADAS-cog and CDR-SOB. At the end of the trial, cognitive functions in the saffron-treated group were significantly improved compared to the placebo group and the frequency of adverse effects between the two groups was similar.

There was also a trial that investigated the efficacy and safety of saffron for the treatment of severe Alzheimer's. In a randomized, double-blind, parallel group trial, saffron was compared with memantine (Farokhnia et al., 2014). In this study, 68 patients with moderate-to-severe AD received memantine (20 mg day⁻¹) or saffron extract (30 mg day⁻¹) capsules for 12 months. Participants were evaluated every month by the severe cognitive impairment rating scale and functional assessment staging along with recording of probable adverse events. Results revealed that the efficacy of saffron was comparable with memantine in decreasing cognitive dysfunction in patients with moderate-to-severe AD. There were no significant differences in adverse effects between the two treatment groups.

28.3 Other herbal medicines and Alzheimer's disease

The concept of herbal medicine is variable and includes simple herbal supplementary agents and therapeutic agents in the traditional medicinal system. Studies evaluating the efficacy of herbal medicine have found solid evidence for its use in the treatment of AD. In addition, research investigating the anti-AD effects of herbal medicine in experimental models are growing. *Ginkgo biloba* leaf, also called ginkgo and *Panax ginseng* root, commonly called ginseng, are the most famous and top-selling herbal supplements for their antioxidant and/or memory-enhancing properties. However, evaluation of their preventive or therapeutic effect in AD patients does not seem to have reached an end (Geun Kim and Sook Oh, 2012).

28.4 Conclusion

AD is one of the leading causes of dementia in the world and its progression involves many factors including cholinesterase activity and A β and tau proteins. Current therapies with AChE inhibitors and NMDA glutamate receptor blockers are only useful for delaying or preventing symptoms of AD for a limited time. Research targeting A β and tau proteins is limited and its efficacy in improving cognition has not been validated in human trials. Natural alternatives that exhibit beneficial effects via multiple targets and pathways could be valuable options. According to the studies presented in this chapter, saffron and particularly crocin have positive effects on the CNS, and crocin, in particular, appears to be multifunctional in protecting brain cells, modulating aggregation of A β and tau proteins, and attenuating cognitive and memory impairment. Although the number of saffron clinical studies in treatment of AD is limited, the results are promising. The findings from animal studies of crocin as well as the clinical studies of saffron confirm crocin as an effective natural alternative in the prevention and treatment of AD.

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